

This is the manual for the anisotropic inversion using FEMTIC.

- The difference from the isotropic inversion is highlighted with a yellow background.
- Important parameters are shown in red.

How to use FEMTIC

1. Serial calculation

- 1) Copy all input files to a work directory.
- 2) Write '1' below 'NUM_THREADS' in 'control.dat'.
- 3) In the work directory, execute the following command.
femtic

2. Parallel calculation only using MPI

- 1) Copy all input files to a work directory.
- 2) Write '1' below 'NUM_THREADS' in 'control.dat'.
- 3) In the work directory, execute the following command.
mpiexec -n [Number of MPI processes] femtic

3. Parallel calculation only using OpenMP

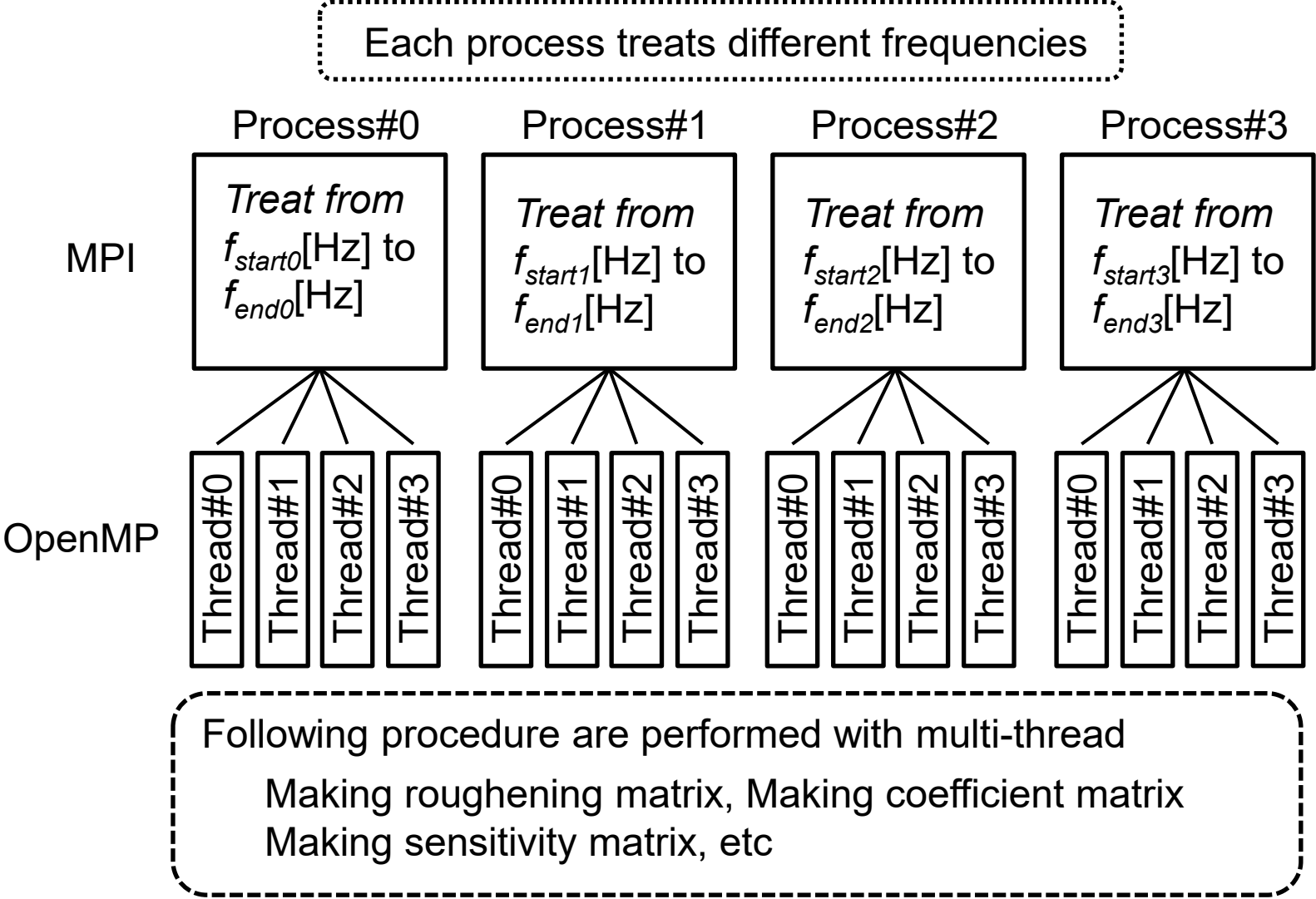
- 1) Copy all input files to a work directory.
- 2) Write number of threads below 'NUM_THREADS' in 'control.dat'.
< Example >
NUM_THREADS
4
- 3) In the work directory, execute the following command.
femtic

4. Hybrid MPI/OpenMP parallel calculation

- 1) Copy all input files to a work directory.
- 2) Write number of threads below 'NUM_THREADS' in 'control.dat'.
< Example >
NUM_THREADS
4
- 1) In the work directory, execute the following command.
mpiexec -n [Number of MPI processes] femtic

OpenMP-MPI hybrid parallelization

OpenMP-MPI Hybrid parallelization speed up the forward and inversion calculation.



Input files

File name	Content
control.dat	Parameters controlling FEMTIC
mesh.dat	Data of computational mesh
resistivity_block_iter[Iter#].dat	Data of parameter cells and initial resistivity values
observe.dat	Observation data
distortion_iter[Iter#].dat	Initial parameters of galvanic distortion

In the file names, [Iter#] indicates the start number of the iteration.
The start number of the iteration can be specified in 'ITERATION' data in 'control.dat'.

File format of control.dat (1/7)

This file contains the parameters controlling FEMTIC.

Keyword	Content	Data type	Option	Default	Example
MESH_TYPE	Type of element	Integer (1 or 2)	1: Tetrahedral element 2: Deformed non-conforming hexahedral element	0	MESH_TYPE 2
INV_METHOD	Type of inversion method	Integer (1)	1: Data space method	0	INV_METHOD 1
NUM_THREADS	Number of threads	Integer greater than or equal to 1		1	NUM_THREADS 4
FWD_SOLVER	Mode of direct sparse solver ^{*1)}	Integer (0, 1 or 2)	0: In-core mode 1: Out-of-core mode is selected when the amount of the memory required by the direct solver is larger than that specified below 'MEM_LIMIT'; otherwise in-core mode is selected. 2: Out-of-core mode.	0	FWD_SOLVER 0
ANISOTROPY	Keyword for activating the anisotropic inversion				ANISOTROPY

*1) In the in-core mode, all information required by the direct solver is stored in memory. On the other hand, in the out-of-core mode, some of the information are stored to and read from temporary data files at hard disk. Computational speed of the in-core mode is much higher than that of the out-of-core mode. However, the in-core mode requires much more memory than the out-of-core mode.

Type of element

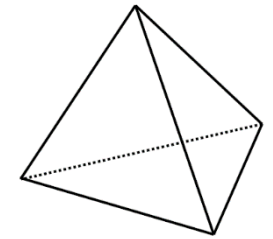
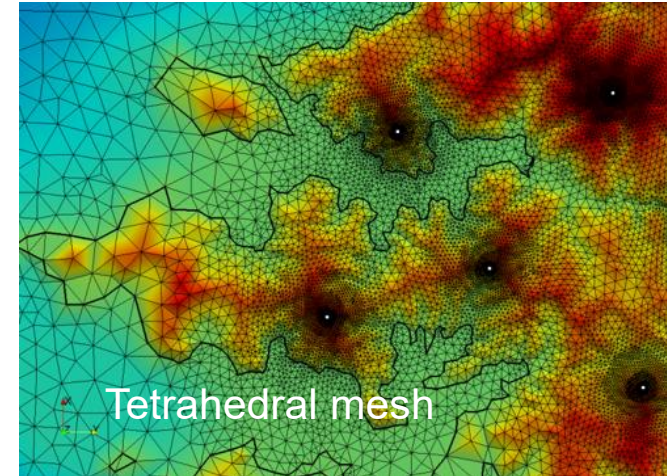
1. Tetrahedral element

2. Non-conforming deformed hexahedral mesh

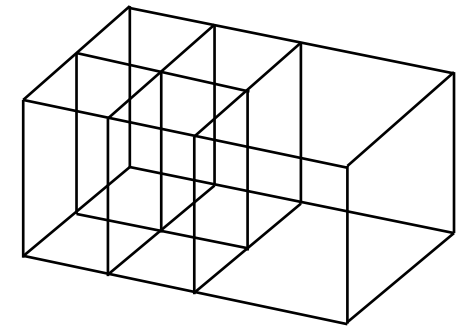
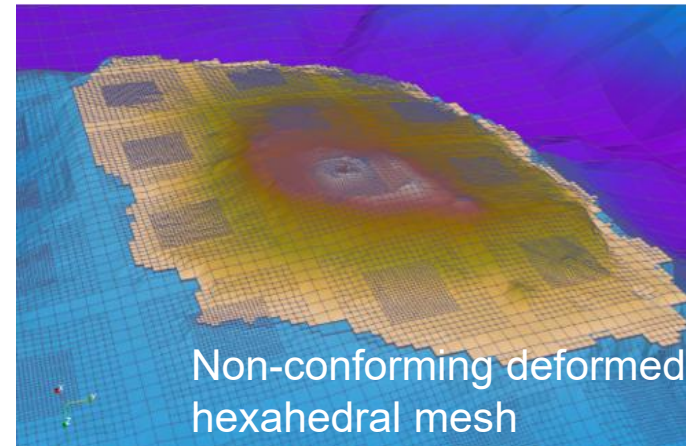
Discretization method of non-conforming deformed hexahedral mesh is based on that of Grayver & Bürg (2014).

In FEMTIC, the division number of one side can be double compared to the other side only in the horizontal direction.

Grayver, A. V., & Bürg, M. (2014). *Robust and scalable 3-D geo-electromagnetic modelling approach using the finite element method*. *Geophysical Journal International*.
<https://doi.org/10.1093/gji/ggu119>



Tetrahedral
element



Non-conforming deformed
hexahedral element

File format of control.dat (2/7)

Keyword	Content	Data type	Option	Default	Example
MEM_LIMIT	The maximum amount of memory (in MB) for direct sparse solver	Real number greater than 0		3000(MB)	MEM_LIMIT 3000
DIV_NUM_RHS_FWD	Division number of the right-hand-side vectors of the linear equation solved for the calculation of sensitivity matrix *1)	Integer greater than or equal to 1		1	DIV_NUM_RHS_FWD 1
DIV_NUM_RHS_INV	Division number of the right-hand-side vectors of the linear equation solved for the calculation of the updates of model parameters. This option is used only when the data-space method is selected *2)	Integer greater than or equal to 1		1	DIV_NUM_RHS_INV 1

*1) In calculating sensitivity matrix, a linear equation with multiple right-hand-sides is solved. When division number of the right-hand-side vectors is one, a routine of PARDISO is called one time to solve the linear equation. On the other hand, when the division number is more than one, that routine is called the specified times and the number of right-hand-side vectors at each time is inversely proportional to the division number as shown in the next slide. In general, the smaller the division number becomes, the faster the speed for solving a linear equation becomes. However, the smaller the division number is, the more memory is required.

*2) When the data-space method is used, a linear equation with multiple right-hand-sides is solved for the calculation of the updates of model parameters. When division number of the right-hand-side vectors is one, a routine of PARDISO is called one time to solve the linear equation. On the other hand, when the division number is more than one, that routine is called the specified times and the number of right-hand-side vectors at each time is inversely proportional to the division number as shown in the next slide. In general, the smaller the division number becomes, the faster the speed for solving a linear equation becomes. However, the smaller the division number is, the more memory is required.

Linear equation to be solved

$$\begin{bmatrix} A \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & x_7 & x_8 \end{bmatrix} = \begin{bmatrix} b_1 & b_2 & b_3 & b_4 & b_5 & b_6 & b_7 & b_8 \end{bmatrix}$$

(1) Division number of the right-hand-side vectors is one

(2) Division number of the right-hand-side vectors is two

Thread #0

Thread #1

$$\begin{bmatrix} A \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix} = \begin{bmatrix} b_1 & b_2 & b_3 & b_4 \end{bmatrix}$$

$$\begin{bmatrix} A \end{bmatrix} \begin{bmatrix} x_5 & x_6 & x_7 & x_8 \end{bmatrix} = \begin{bmatrix} b_5 & b_6 & b_7 & b_8 \end{bmatrix}$$

Thread #0

Thread #1

$$\begin{bmatrix} A \end{bmatrix} \begin{bmatrix} x_1 & x_2 \end{bmatrix} = \begin{bmatrix} b_1 & b_2 \end{bmatrix}$$

$$\begin{bmatrix} A \end{bmatrix} \begin{bmatrix} x_3 & x_4 \end{bmatrix} = \begin{bmatrix} b_3 & b_4 \end{bmatrix}$$

A routine of PARDISO is called again



Thread #0

Thread #1

$$\begin{bmatrix} A \end{bmatrix} \begin{bmatrix} x_5 & x_6 \end{bmatrix} = \begin{bmatrix} b_5 & b_6 \end{bmatrix}$$

$$\begin{bmatrix} A \end{bmatrix} \begin{bmatrix} x_7 & x_8 \end{bmatrix} = \begin{bmatrix} b_7 & b_8 \end{bmatrix}$$

File format of control.dat (3/7)

Keyword	Content	Data type	Option	Default	Example
ELEC_FIELD	Type of the electric field used to calculate response functions	Integer (0, 1 or -1)	0: Horizontal electric field 1: Tangential electric field -1: Type of the electric field of each station is individually selected *2)	0	ELEC_FIELD 0
OWNER_ELEMENT	Type of owner element *1) of observation stations	Integer (0, 1 or -1)	0: Downward element 1: Upward element -1: Type of owner element of each station is individually selected *2)	0	OWNER_ELEMENT 0
SMALL_VALUE	A small positive value used to regularize roughening matrix (Usui et al., 2017). This option is necessary only when the data-space method is selected.	Positive real number			SMALL_VALUE 1.0e-4

*1) Owner element denotes the element in which the electric field and magnetic field of the observation station are interpolated.

*2) When this option is selected, you need to specify the type of the electric field and/or the owner element for each station in 'observe.dat'.

File format of control.dat (4/7)

Keyword	Content	Data type	Option	Default	Example
OUTPUT_PARAM	Number of output variables	Integer greater than or equal to 0			OUTPUT_PARAM 2 0 4
	Output variables for the visualization with ParaView ^{*1)}	Integer(s)			
OFFILE_TYPE	Format of output file for ParaView	Integer (0 or 1)	0: VTK file format (ASCII) 1: Ensight Gold file format (binary)	1	OFFILE_TYPE 1
MOVE_OBS_LOC	When you write this keyword, observation station is forced to be located at the center of the element face				MOVE_OBS_LOC
DISTORTION	Treatment of galvanic distortion	Integer (0, 1, 2 or 3)	0: Distortion matrix is not estimated 1: Differences of distortion matrix and unit matrix are estimated. 2: Both gains and rotations are estimated ^{*2)} . 3: Only gains are estimated.	None. You must write this keyword and an option.	DISTORTION 1

*1) Detail of this option is described in the next slide.

*2) Distortion matrix is decomposed as Smith (1995)

Smith, J. T. (1995). *Understanding telluric distortion matrices*. *Geophysical Journal International*, 122(1), 219–226.

<https://doi.org/10.1111/j.1365-246X.1995.tb03549.x>

Format about the keyword 'OUTPUT_PARAM'

Under the keyword, you write the number of output variables first.

Following the number, you need to write the index numbers of the variables needed.

The variables you selected are outputted by VTK file format (ASCII) or Enight Gold file format (binary) depending on the setting of 'OFFILE_TYPE'.

The output files can be read and visualized by ParaView.

<Example>

By the following setting, the resistivity and sensitivity are only outputted.

```
OUTPUT_PARAM
```

```
2
```

```
0 4
```

Output file names of Enight Gold file format

Index	Variable type	Output file name
0	Resistivity	RhoXX.iter[Iter#], RhoYY.iter[Iter#], RhoZZ.iter[Iter#], Strike.iter[Iter#], Dip.iter[Iter#], Slant.iter[Iter#], Anisotropy.iter[Iter#]
1	Electric field	ReE_Freq[Freq#]_ExPol.iter[Iter#] ImE_Freq[Freq#]_ExPol.iter[Iter#] ReE_Freq[Freq#]_EyPol.iter[Iter#] ImE_Freq[Freq#]_EyPol.iter[Iter#]
2	Magnetic field	ReH_Freq[Freq#]_ExPol.iter[Iter#] ImH_Freq[Freq#]_ExPol.iter[Iter#] ReH_Freq[Freq#]_EyPol.iter[Iter#] ImH_Freq[Freq#]_EyPol.iter[Iter#]
3	Electric current	Rej_Freq[Freq#]_ExPol.iter[Iter#] Imj_Freq[Freq#]_ExPol.iter[Iter#] Rej_Freq[Freq#]_EyPol.iter[Iter#] Imj_Freq[Freq#]_EyPol.iter[Iter#]
4	Sensitivity	Sensitivity.iter[Iter#] NormalizedSensitivity.iter[Iter#]
5	Sensitivity density	SensitivityDensity.iter[Iter#] NormalizedSensitivityDensity.iter[Iter#]

*1) When you select VTK file format (ASCII) as output file format, output variables are written in the files named as 'result_[PE#]_iter[Iter#].vtk'.

*2) In the file names, [PE#] and [Iter#] indicate process number and iteration number, respectively.

File format of control.dat (5/7)

Keyword	Content	Data type	Option	Default	Example
TRADE_OFF_PARAM	Trade-off parameters	Real positive value(s)	The number of option(s) depends on the option of the keyword 'DISTORTION'. If the option of 'DISTORTION' is '0', α_1, α_2, and α_3 should be specified If the option of 'DISTORTION' is '1' or '3', α_1, α_2, α_3, and β should be specified. If the option of 'DISTORTION' is '2', α_1, α_2, α_3, β_1 and β_2 should be specified.	None. You must write this keyword and option(s).	TRADE_OFF_PARAM 1.0 0.1 100.0
ITERATION	Start and end iteration number	Two integers greater than or equal to 0	Start and end iteration number	None. You must write this keyword and options.	ITERATION 0 10
CONVERGE	Convergence threshold (%) of Gauss-Newton iteration	Real value greater than 0	If the change rates of the objective function and its respective terms of the current iteration from those of the previous iteration are less than the threshold value, the Gauss-Newton iteration is finished.	1.0	CONVERGE 1.0

Objective function

The form of objective function depends on the setting of 'DISTORTION'.

0: Distortion matrix is not estimated

$$\phi(\mathbf{m}) = \phi_d(\mathbf{m}) + \alpha_1^2 \phi_r(\mathbf{m}) + \alpha_2^2 \phi_a(\mathbf{m}) + \alpha_3^2 \phi_c(\mathbf{m})$$

$\phi_d(\mathbf{m})$: Data misfit

$\phi_m(\mathbf{m})$: Roughness

$\phi_a(\mathbf{m})$: Anisotropy

$\phi_c(\mathbf{m})$: Cross product

$\alpha_1, \alpha_2, \alpha_3$: Trade-off parameters

1: Differences of distortion matrix $\mathbf{C}$ and unit matrix are estimated

$$\phi(\mathbf{m}) = \phi_d(\mathbf{m}) + \alpha_1^2 \phi_r(\mathbf{m}) + \alpha_2^2 \phi_a(\mathbf{m}) + \alpha_3^2 \phi_c(\mathbf{m}) + \beta^2 \phi_c(\mathbf{m})$$

$$\phi_c(\mathbf{m}) = \sum_{i=1}^{\#sites} \left[|c'_{xx}|^2 + |c'_{xy}|^2 + |c'_{yx}|^2 + |c'_{yy}|^2 \right]$$

β : Trade-off parameter for distortion matrix

$$\mathbf{C}' = \mathbf{C} - \mathbf{I} = \begin{pmatrix} c_{xx} & c_{xy} \\ c_{yx} & c_{yy} \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} c'_{xx} & c'_{xy} \\ c'_{yx} & c'_{yy} \end{pmatrix}$$

2: Both gains and rotations are estimated

$$\phi(\mathbf{m}) = \phi_d(\mathbf{m}) + \alpha_1^2 \phi_r(\mathbf{m}) + \alpha_2^2 \phi_a(\mathbf{m}) + \alpha_3^2 \phi_c(\mathbf{m}) + \beta_1^2 \phi_{c1}(\mathbf{m}) + \beta_2^2 \phi_{c2}(\mathbf{m})$$

$$\phi_{c1}(\mathbf{m}) = \sum_{i=1}^{\#site} [G_x^2 + G_y^2], \phi_{c2}(\mathbf{m}) = \sum_{i=1}^{\#site} [\beta_x^2 + \beta_y^2]$$

β_1, β_2 : Trade-off parameters for gains and rotations

$$\mathbf{C} = \begin{pmatrix} 10^{G_x} \begin{pmatrix} \cos \beta_x \\ \sin \beta_x \end{pmatrix} & 10^{G_y} \begin{pmatrix} -\sin \beta_y \\ \cos \beta_y \end{pmatrix} \end{pmatrix}$$

3: Only gains are estimated.

$$\phi(\mathbf{m}) = \phi_d(\mathbf{m}) + \alpha_1^2 \phi_r(\mathbf{m}) + \alpha_2^2 \phi_a(\mathbf{m}) + \alpha_3^2 \phi_c(\mathbf{m}) + \beta^2 \phi_c(\mathbf{m})$$

$$\phi_c(\mathbf{m}) = \sum_{i=1}^{\#site} [G_x^2 + G_y^2]$$

β : Trade-off parameter for gains

$$\mathbf{C} = \begin{pmatrix} 10^{G_x} & 0 \\ 0 & 10^{G_y} \end{pmatrix}$$

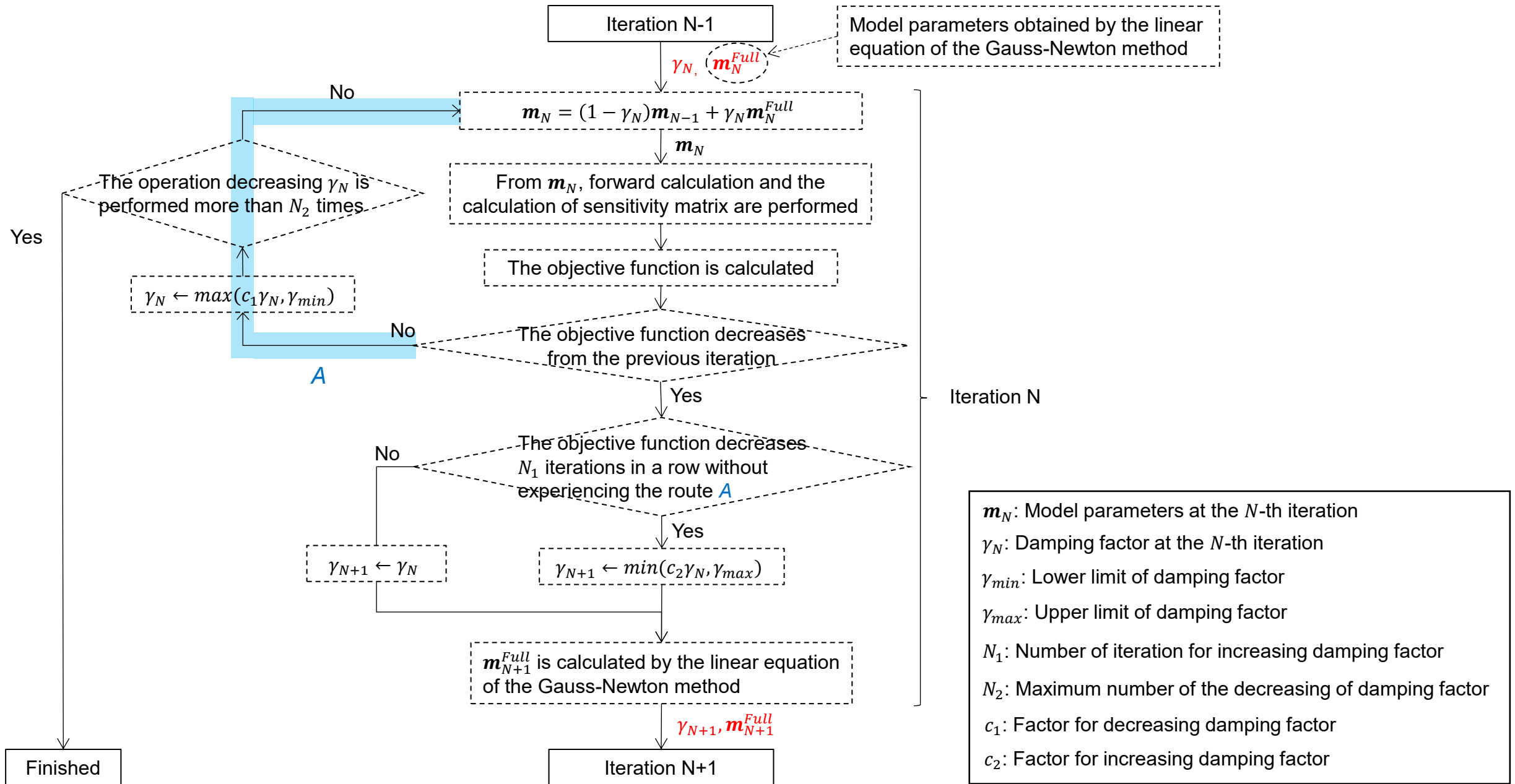
File format of control.dat (6/7)

Keyword	Content	Data type	Option	Default	Example
DECREASE_THRESHOLD	Threshold value for determining whether the objective function decreases. If the difference of the objective function of the current iteration and the previous iteration is less than the threshold value, the objective function is deemed to decrease.	Positive real value		0.001	DECREASE_THRESHOLD 0.001
STEP_LENGTH ^{*1)}	Initial damping factor	Positive real value		0.50	STEP_LENGTH 0.5 0.1 0.5 3 0.5 1.2
	Lower limit of damping factor	Positive real value		0.10	
	Upper limit of damping factor	Positive real value		1.00	
	Number of iteration for increasing damping factor	Positive integer		3	
	Factor for decreasing damping factor	Real value less than or equal to 1.0		0.50	
	Factor for increasing damping factor	Real value greater than or equal to 1.0		1.25	
RETRIAL	Maximum number of the decreasing of damping factor	Positive integer		5	RETRIAL 3

*1) Modification algorithm of the damping factor is shown as flow chart in the next slide.

File format of control.dat (7/7)

Keyword	Content	Data type	Option	Default	Example
APP_PHS_OPTION	Option about treatment of apparent resistivity and phase	Integer (0 or 1)	0:No special treatment 1:Impedance tensor is used instead of apparent resistivity and phase if the sign of the real part of impedance tensor component is different between observed and calculated responses. This treatment will improve the stabilization of the inversion using apparent resistivity and phase.	0	APP_PHS_OPTION 1
SENSE_MAT_DIRECTORY	Directory where out-of-core files for the sensitivity matrix will be outputted	string		Directory where input files are stored	SENSE_MAT_DIRECTORY /work/temp
END	Indication of the end of controlling parameters				END



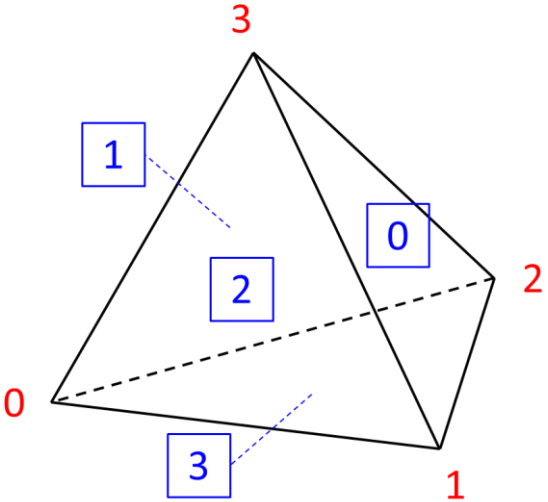
Recommended setting for initial trial

```
MESH_TYPE
2
OUTPUT_PARAM
1
0
DIV_NUM_RHS_FWD
10
INV_METHOD
1
SMALL_VALUE
1.0e-4
ANISOTROPY
DISTORTION
0
TRADE_OFF_PARAM
10.0 1.0 100.0
ITERATION
0 20
STEP_LENGTH
0.5 0.1 0.8
2
0.6 1.2
END
```

File format of mesh.dat (for tetrahedral elements) (1/3)

TETRA			
Number of nodes (N_n)			
0	X coordinate value of the 1 st node	Y coordinate value of the 1 st node	Z coordinate value of the 1 st node
⋮			
$N_n - 1$	X coordinate value of the N_n -th node	Y coordinate value of the N_n -th node	Z coordinate value of the N_n -th node

Number of elements (N_e)				
0	Index of the adjacent element sharing face 0		Index of the adjacent element sharing face 1	
	Index of node 0	Index of node 1	Index of node 2	Index of node 3
⋮				
$N_e - 1$	Index of the adjacent element sharing face 0		Index of the adjacent element sharing face 1	
	Index of node 0	Index of node 1	Index of node 2	Index of node 3



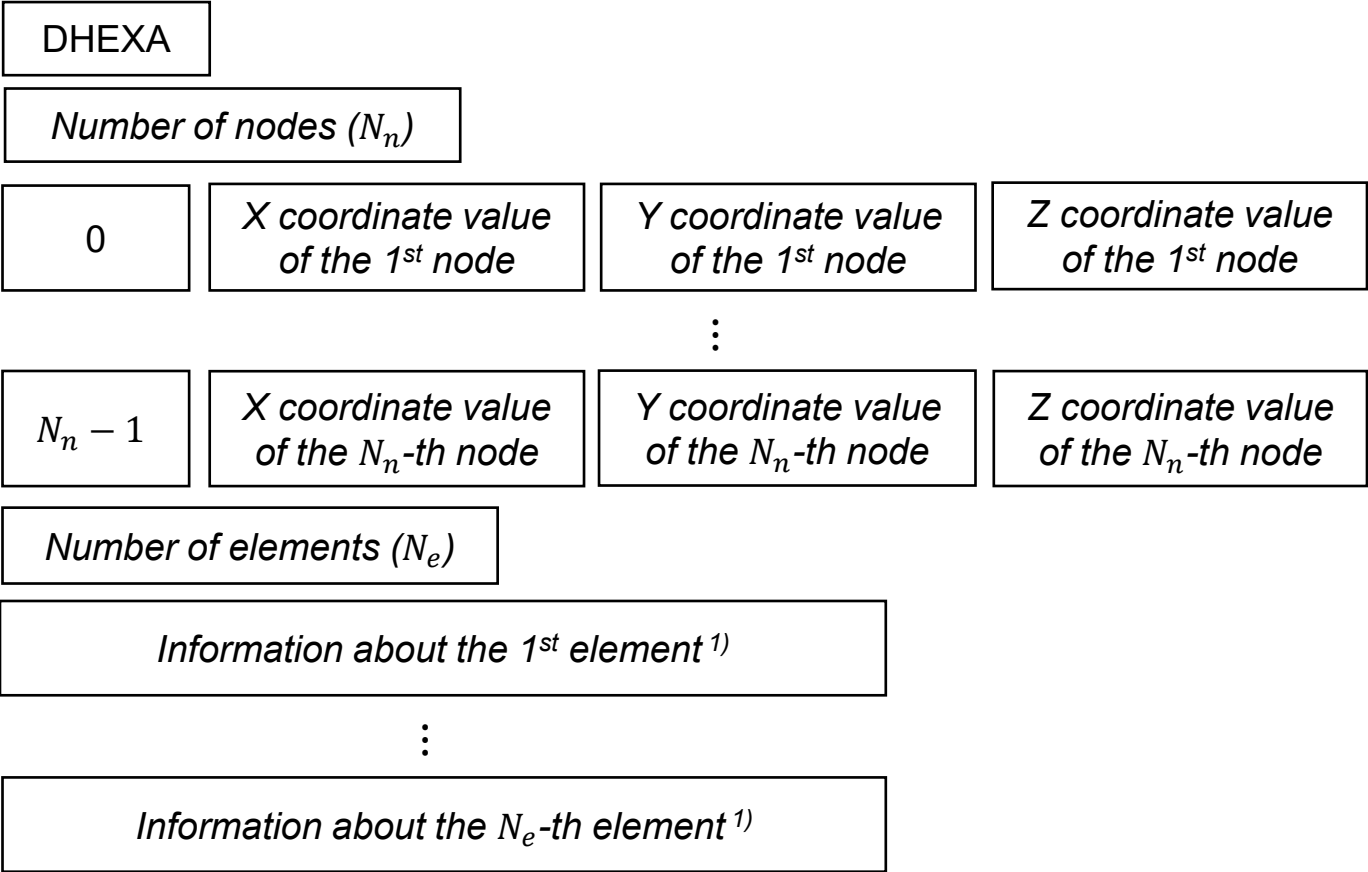
File format of mesh.dat (for tetrahedral elements) (2/3)

Number of the element faces on the Y-Z plane (-X side) (N_{-YZ})	
Element index of the 1 st element face	Face index of the 1 st element face
⋮	
Element index of the N_{-YZ} -th element face	Face index of the N_{-YZ} -th element face
Number of the element faces on the Y-Z plane (+X side) (N_{+YZ})	
Element index of the 1 st element face	Face index of the 1 st element face
⋮	
Element index of the N_{+YZ} -th element face	Face index of the N_{+YZ} -th element face
Number of the element faces on the Z-X plane (-Y side) (N_{-ZX})	
Element index of the 1 st element face	Face index of the 1 st element face
⋮	
Element index of the N_{-ZX} -th element face	Face index of the N_{-ZX} -th element face
Number of the element faces on the Z-X plane (+Y side) (N_{+ZX})	
Element index of the 1 st element face	Face index of the 1 st element face
⋮	
Element index of the N_{+ZX} -th element face	Face index of the N_{+ZX} -th element face

File format of mesh.dat (for tetrahedral elements) (3/3)

Number of the element faces on the X-Y plane (-Z side) (N_{-XY})	
Element index of the 1 st element face	Face index of the 1 st element face
⋮	
Element index of the N_{-XY} -th element face	Face index of the N_{-XY} -th element face
Number of the element faces on the X-Y plane (+Z side) (N_{+XY})	
Element index of the 1 st element face	Face index of the 1 st element face
⋮	
Element index of the N_{+XY} -th element face	Face index of the N_{+XY} -th element face
Number of the element faces on the Earth's surface (N_{surf})	
Element index of the 1 st element face	Face index of the 1 st element face
⋮	
Element index of the N_{surf} -th element face	Face index of the N_{surf} -th element face

File format of mesh.dat (for deformed non-conforming hexahedral elements) (1/4)



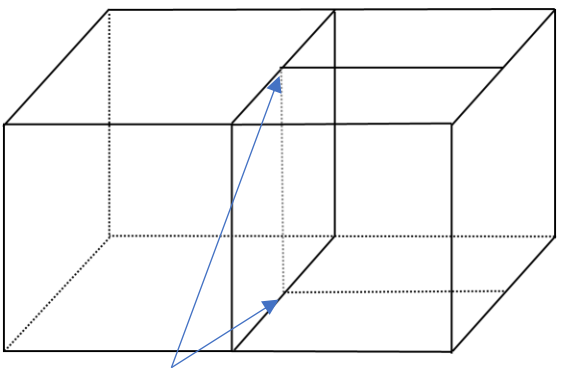
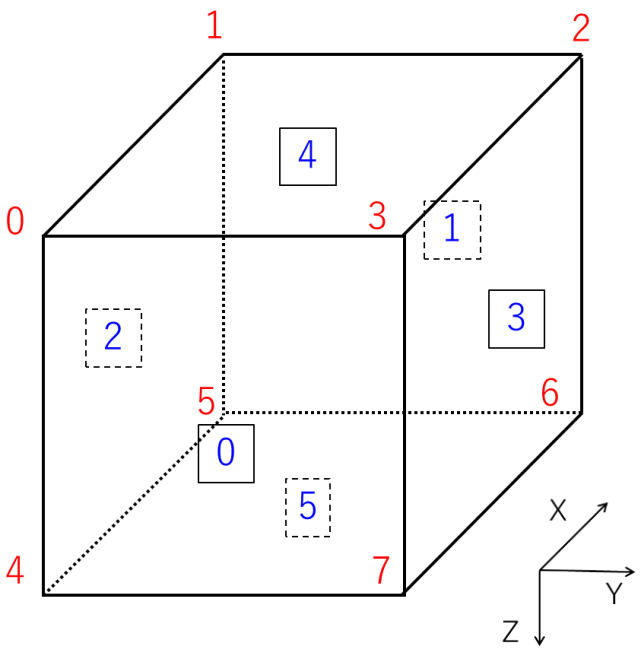
*1) The format of the information about element is described in the next slide

File format of mesh.dat (for deformed non-conforming hexahedral elements) (2/4)

The format of the information about element

Element Index	Index of node 0	Index of node 1	Index of node 2	Index of node 3	Index of node 4	Index of node 5	Index of node 6	Index of node 7
Total number of the adjacent elements sharing face 0		The element index of the 1st adjacent element		...		The element index of the last adjacent element		
Total number of the adjacent elements sharing face 1		The element index of the 1st adjacent element		...		The element index of the last adjacent element		
Total number of the adjacent elements sharing face 2		The element index of the 1st adjacent element		...		The element index of the last adjacent element		
Total number of the adjacent elements sharing face 3		The element index of the 1st adjacent element		...		The element index of the last adjacent element		
Total number of the adjacent elements sharing face 4		The element index of the 1st adjacent element		...		The element index of the last adjacent element		
Total number of the adjacent elements sharing face 5		The element index of the 1st adjacent element		...		The element index of the last adjacent element		

Total number of the adjacent elements is larger than 1 if there are hanging nodes on the face. If there is no adjacent element, that can occur at the outer boundary of the mesh, total number of the adjacent elements should be zero or a negative value.

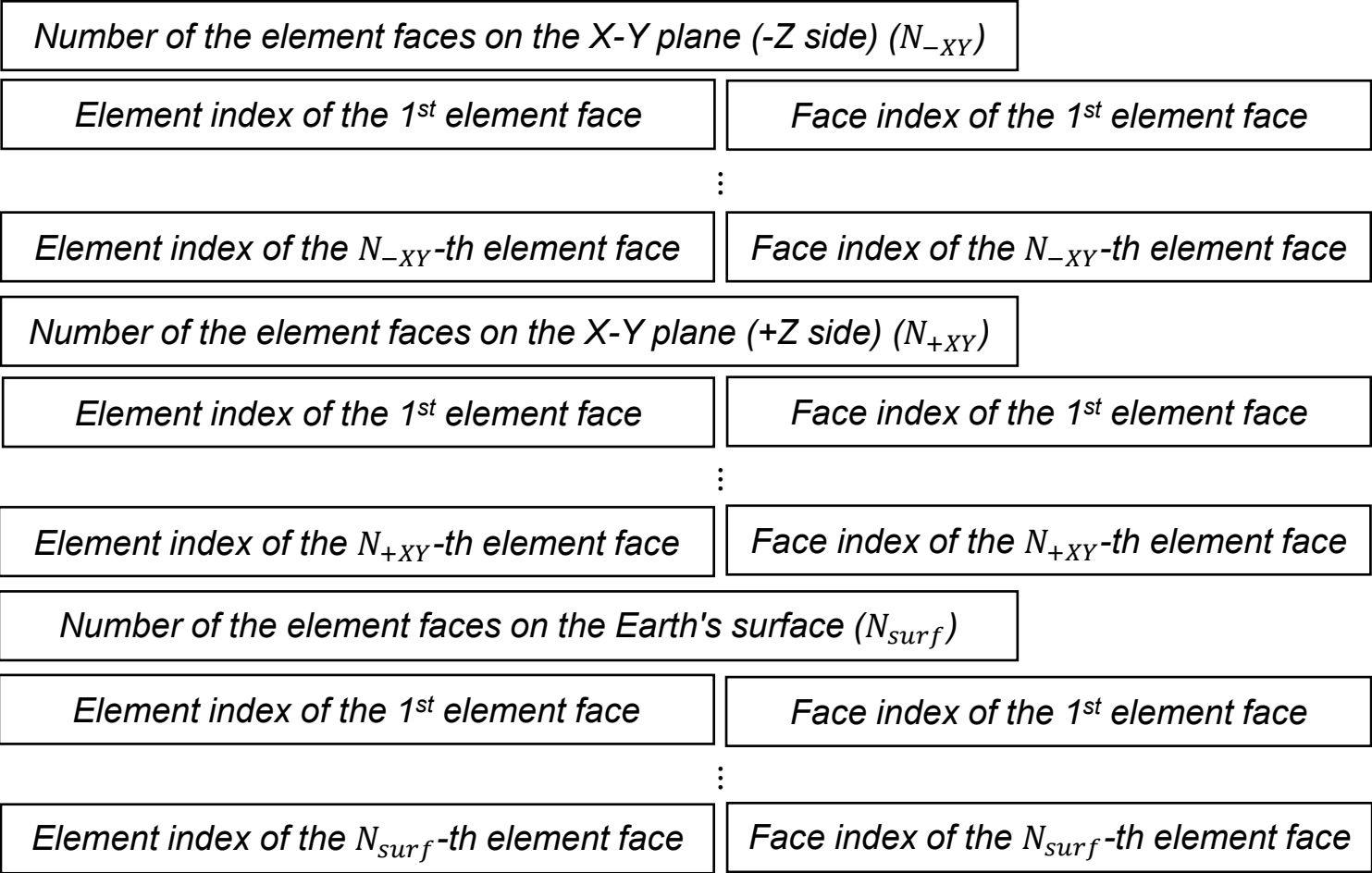


Hanging node

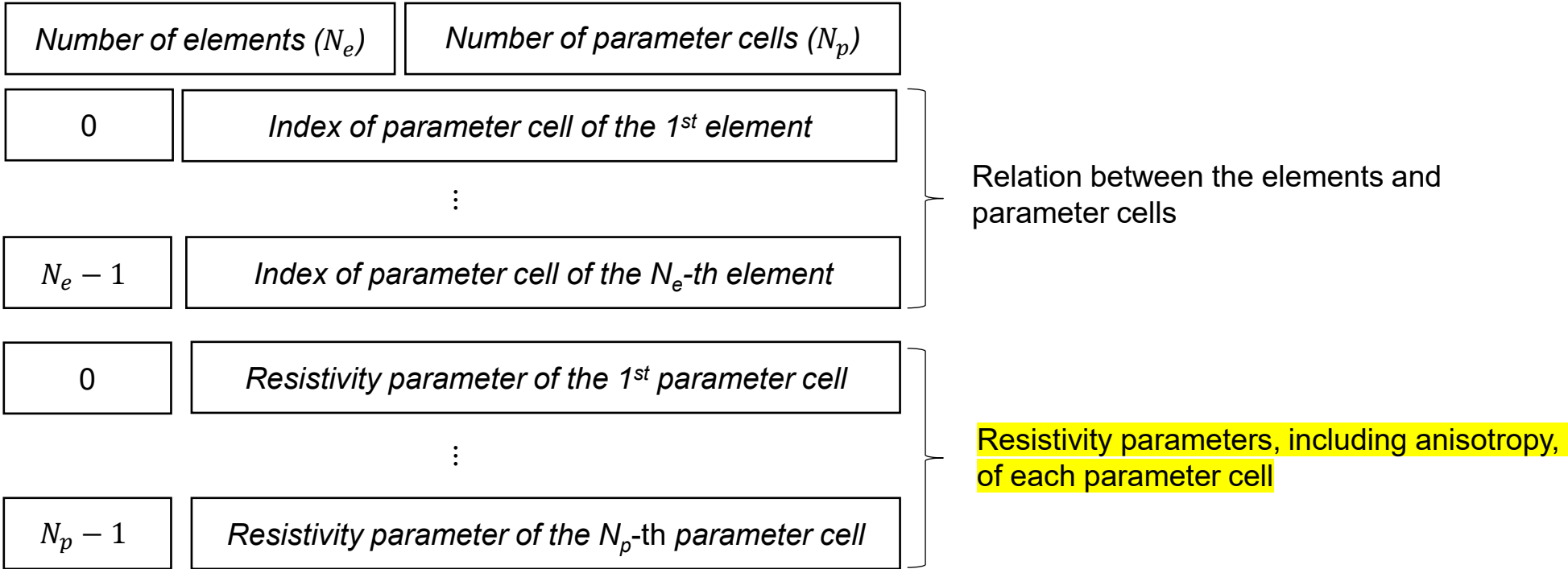
File format of mesh.dat (for deformed non-conforming hexahedral elements) (3/4)

Number of the element faces on the Y-Z plane (-X side) (N_{-YZ})	
Element index of the 1 st element face	Face index of the 1 st element face
⋮	
Element index of the N_{-YZ} -th element face	Face index of the N_{-YZ} -th element face
Number of the element faces on the Y-Z plane (+X side) (N_{+YZ})	
Element index of the 1 st element face	Face index of the 1 st element face
⋮	
Element index of the N_{+YZ} -th element face	Face index of the N_{+YZ} -th element face
Number of the element faces on the Z-X plane (-Y side) (N_{-ZX})	
Element index of the 1 st element face	Face index of the 1 st element face
⋮	
Element index of the N_{-ZX} -th element face	Face index of the N_{-ZX} -th element face
Number of the element faces on the Z-X plane (+Y side) (N_{+ZX})	
Element index of the 1 st element face	Face index of the 1 st element face
⋮	
Element index of the N_{+ZX} -th element face	Face index of the N_{+ZX} -th element face

File format of mesh.dat (for deformed non-conforming hexahedral elements) (4/4)



resistivity_block_iter[lter#].dat (1/3)



The format for the resistivity parameters is shown in the next slide

resistivity_block_iter[lter#].dat (2/3)

The format for the resistivity parameters

For a parameter cell with **isotropic** resistivity,

Parameter cell index	0	Resistivity (Ωm)	Lower limit of resistivity (Ωm)	Upper limit of resistivity (Ωm)	Fixing resistivity or not
0: Resistivity is NOT fixed 1: Resistivity is fixed					

For a parameter cell with **transverse isotropic** resistivity,

Parameter cell index	1	ρ_{xx} (Ωm)	ρ_{yy} (Ωm)	θ_S (deg.)	θ_D (deg.)	Lower limit of resistivity (Ωm)	Upper limit of resistivity (Ωm)
Fixing ρ_{xx} or not		Fixing ρ_{yy} or not		Fixing θ_S or not		Fixing θ_D or not	
0: Principal resistivity or rotation angle is NOT fixed 1: Principal resistivity or rotation angle is fixed							

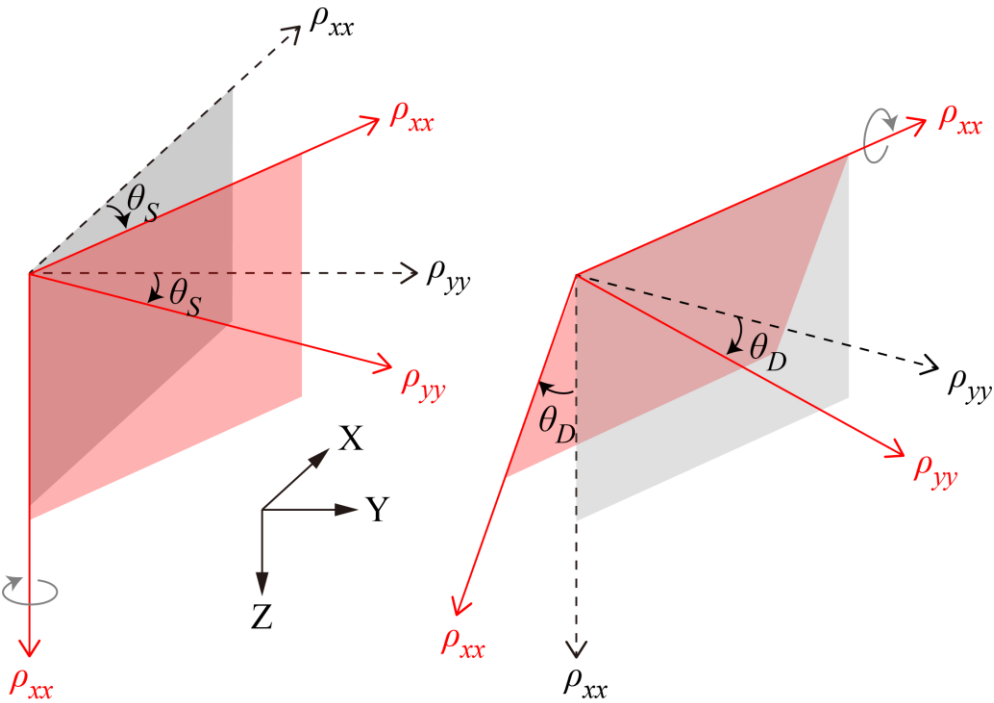
For a parameter cell with **generally anisotropic** resistivity,

Parameter cell index	2	ρ_{xx} (Ωm)	ρ_{yy} (Ωm)	ρ_{zz} (Ωm)	θ_S (deg.)	θ_D (deg.)	θ_L (deg.)
		Lower limit of resistivity (Ωm)	Upper limit of resistivity (Ωm)	Fixing ρ_{xx} or not	Fixing ρ_{yy} or not	Fixing ρ_{zz} or not	
		Fixing θ_S or not	Fixing θ_D or not	Fixing θ_L or not	0: Principal resistivity or rotation angle is NOT fixed 1: Principal resistivity or rotation angle is fixed		

The second column is the type of anisotropy/isotropy.

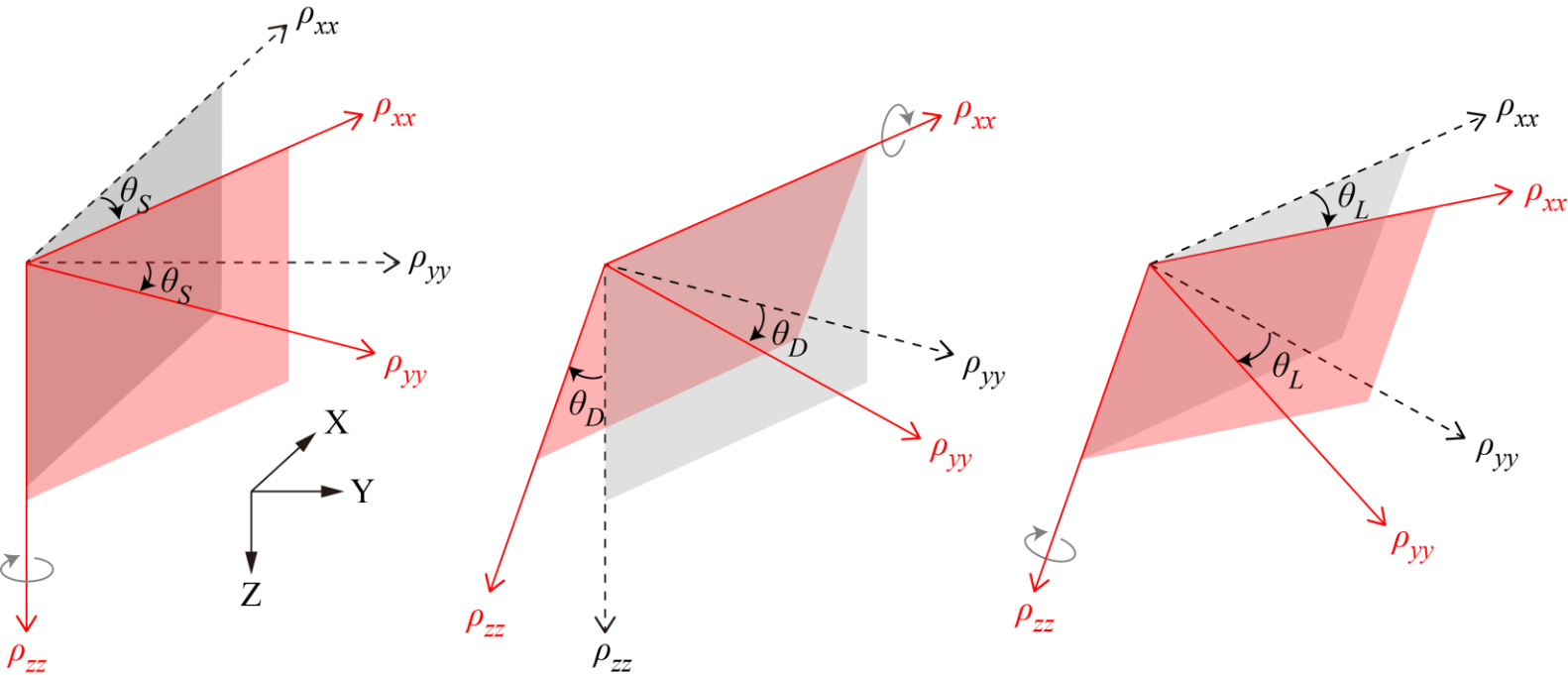
General anisotropy is supported only for the forward calculation, i.e., when the start and end iteration numbers are the same.

resistivity_block_iter[lter#].dat (3/3)



Transverse isotropic resistivity

Generally anisotropic resistivity



File format of observe.dat (1/2)

The following response functions can be used as the observed data.

- *Impedance tensor*

$$\begin{pmatrix} E_x \\ E_y \end{pmatrix} = \begin{pmatrix} Z_{xx} & Z_{xy} \\ Z_{yx} & Z_{yy} \end{pmatrix} \begin{pmatrix} H_x \\ H_y \end{pmatrix}$$

- *Vertical magnetic transfer function (Tipper)*

$$H_z = \begin{pmatrix} T_{zx} & T_{zy} \end{pmatrix} \begin{pmatrix} H_x \\ H_y \end{pmatrix}$$

- *Horizontal magnetic transfer function between two sites*

$$\begin{pmatrix} H_x^{Site1} \\ H_y^{Site1} \end{pmatrix} = \begin{pmatrix} T_{xx} & T_{xy} \\ T_{yx} & T_{yy} \end{pmatrix} \begin{pmatrix} H_x^{Site2} \\ H_y^{Site2} \end{pmatrix}$$

- *Phase tensor (Caldwell et al. 2004)*

$$\begin{pmatrix} \Phi_{11} & \Phi_{12} \\ \Phi_{21} & \Phi_{22} \end{pmatrix} = \begin{pmatrix} Re(Z_{xx}) & Re(Z_{xy}) \\ Re(Z_{yx}) & Re(Z_{yy}) \end{pmatrix}^{-1} \begin{pmatrix} Im(Z_{xx}) & Im(Z_{xy}) \\ Im(Z_{yx}) & Im(Z_{yy}) \end{pmatrix}$$

- *Apparent resistivity*

$$\rho_{a,ij} = \frac{1}{\omega\mu_0} |Z_{ij}|^2$$

- *Phase*

$$\theta_{ij} = \begin{cases} \text{Arctan}\left(\frac{Im(Z_{ij})}{Re(Z_{ij})}\right) & (Re(Z_{ij}) \geq 0) \\ \text{Arctan}\left(\frac{Im(Z_{ij})}{Re(Z_{ij})}\right) + \pi & (Re(Z_{ij}) < 0 \text{ and } Im(Z_{ij}) \geq 0) \\ \text{Arctan}\left(\frac{Im(Z_{ij})}{Re(Z_{ij})}\right) - \pi & (Re(Z_{ij}) < 0 \text{ and } Im(Z_{ij}) < 0) \end{cases}$$

The use of the apparent resistivity and phase is not recommended. The quadrant of the phase changes, particularly for the diagonal components. That can worsen the convergence property of the inversion.

Response functions must be written in SI units with $e^{-i\omega t}$ time dependence.

File format of observe.dat (2/2)

The following response functions can be used as the observed data.

- *Network-MT response function*

$$V = \begin{pmatrix} V_x & V_y \end{pmatrix} \begin{pmatrix} H_x \\ H_y \end{pmatrix}$$

- *Impedance tensor obtained from the Network-MT response functions of two different dipoles*
- *Apparent resistivity and phase obtained from the Network-MT response functions of two different dipoles*

The use of the apparent resistivity and phase is not recommended. The quadrant of the phase changes, particularly for the diagonal components. That can worsen the convergence property of the inversion.

Response functions must be written in SI units with $e^{-i\omega t}$ time dependence.

Impedance tensor (1/2)

MT	Number of observation stations of impedance tensor (N_{imp})				
Station ID	Station ID for horizontal magnetic field	Type of the owner element ^{*1)}	Type of the electric field ^{*2)}	X coordinate value of the station (km)	Y coordinate value of the station (km)
Number of frequencies (N_f)					
Observed data of the 1 st frequency ^{*3)}					
⋮					
Observed data of the N_f -th frequency ^{*3)}					
					Data for the 1 st station of impedance tensor
⋮					
					Data for the N_{imp} -th station of impedance tensor

*1) Type of the owner element (0: Downward element, 1: Upward element) is needed only if the option of the keyword ‘OWNER_ELEMENT’ in ‘control.dat’ is -1.

*2) Type of the electric field (0: Horizontal electric field, 1: Tangential electric field) is needed only if the option of the keyword ‘ELEC_FIELD’ in ‘control.dat’ is -1.

*3) The format of the observed data of each frequency is described in the next slide

Impedance tensor (2/2)

The format of the observed data of each frequency

<i>Frequency(Hz)</i>	<i>Re(Z_{xx})</i> (V/A)	<i>Im(Z_{xx})</i> (V/A)	<i>Re(Z_{xy})</i> (V/A)	<i>Im(Z_{xy})</i> (V/A)	<i>Re(Z_{yx})</i> (V/A)	<i>Im(Z_{yx})</i> (V/A)	<i>Re(Z_{yy})</i> (V/A)	<i>Im(Z_{yy})</i> (V/A)
	<i>SE of Re(Z_{xx})</i>	<i>SE of Im(Z_{xx})</i>	<i>SE of Re(Z_{xy})</i>	<i>SE of Im(Z_{xy})</i>				
		<i>SE of Re(Z_{yx})</i>	<i>SE of Im(Z_{yx})</i>	<i>SE of Re(Z_{yy})</i>	<i>SE of Im(Z_{yy})</i>			

If SE is negative, the corresponding component is not used in the inversion.

Apparent resistivity & phase (1/2)

APP_RES_AND_PHS		Number of observation stations of apparent resistivity & phase (N_{app})			
Station ID	Station ID for horizontal magnetic field	Type of the owner element ^{*1)}	Type of the electric field ^{*2)}	X coordinate value of the station (km)	Y coordinate value of the station (km)
Number of frequencies (N_f)					
Observed data of the 1 st frequency ^{*3)}					
⋮					
Observed data of the N_f -th frequency ^{*3)}					
				Data for the 1 st station of apparent resistivity & phase	
⋮					
				Data for the N_{app} -th station of apparent resistivity & phase	

*1) Type of the owner element (0: Downward element, 1: Upward element) is needed only if the option of the keyword 'OWNER_ELEMENT' in 'control.dat' is -1.

*2) Type of the electric field (0: Horizontal electric field, 1: Tangential electric field) is needed only if the option of the keyword 'ELEC_FIELD' in 'control.dat' is -1.

*3) The format of the observed data of each frequency is described in the next slide

Apparent resistivity & phase (2/2)

The format of the observed data of each frequency

<i>Frequency(Hz)</i>	<i>$\rho_{a,xx}$ (Ωm)</i>	<i>θ_{xx} (deg.)</i>	<i>$\rho_{a,xy}$ (Ωm)</i>	<i>θ_{xy} (deg.)</i>	<i>$\rho_{a,yx}$ (Ωm)</i>	<i>θ_{yx} (deg.)</i>	<i>$\rho_{a,yy}$ (Ωm)</i>	<i>θ_{yy} (deg.)</i>
	<i>SE of $\rho_{a,xx}$</i>	<i>SE of θ_{xx}</i>	<i>SE of $\rho_{a,xy}$</i>	<i>SE of θ_{xy}</i>				
		<i>SE of $\rho_{a,yx}$</i>	<i>SE of θ_{yx}</i>	<i>SE of $\rho_{a,yy}$</i>	<i>SE of θ_{yy}</i>			

If SE is negative, the corresponding component is not used in the inversion.

Vertical magnetic transfer function (1/2)

VTF	Number of observation stations of vertical magnetic transfer function (N_{vtf})			
Station ID	Station ID for horizontal magnetic field	Type of the owner element ^{*1)}	X coordinate value of the station (km)	Y coordinate value of the station (km)
Number of frequencies (N_f)				
Observed data of the 1 st frequency ^{*2)}				
⋮				
Observed data of the N_f -th frequency ^{*2)}				
Data for the 1 st station of vertical magnetic transfer function				
⋮				
Data for the N_{vtf} -th station of vertical magnetic transfer function				

*1) Type of the owner element (0: Downward element, 1: Upward element) is needed only if the option of the keyword 'OWNER_ELEMENT' in 'control.dat' is -1.

*2) The format of the observed data of each frequency is described in the next slide

Vertical magnetic transfer function (2/2)

The format of the observed data of each frequency

<i>Frequency(Hz)</i>	<i>Re(T_{zx})</i>	<i>Im(T_{zx})</i>	<i>Re(T_{zy})</i>	<i>Im(T_{zy})</i>	<i>SE of Re(T_{zx})</i>	<i>SE of Im(T_{zx})</i>	<i>SE of Re(T_{zy})</i>	<i>SE of Im(T_{zy})</i>
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If SE is negative, the corresponding component is not used in the inversion.

Horizontal magnetic transfer function (1/2)

HTF	Number of observation stations of horizontal magnetic transfer function (N_{htf})			
Station ID for the output-side horizontal magnetic field	Station ID for the input-side horizontal magnetic field	Type of the owner element ^{*1)}	X coordinate value of the station (km)	Y coordinate value of the station (km)
Number of frequencies (N_f)				
Observed data of the 1 st frequency ^{*2)}				
⋮				
Observed data of the N_f -th frequency ^{*2)}				
Data for the 1 st station of horizontal magnetic transfer function				
⋮				
Data for the N_{htf} -th station of horizontal magnetic transfer function				

*1) Type of the owner element (0: Downward element, 1: Upward element) is needed only if the option of the keyword ‘OWNER_ELEMENT’ in ‘control.dat’ is -1.

*2) The format of the observed data of each frequency is described in the next slide

Horizontal magnetic transfer function(2/2)

The format of the observed data of each frequency

<i>Frequency(Hz)</i>	<i>Re(T_{xx})</i>	<i>Im(T_{xx})</i>	<i>Re(T_{xy})</i>	<i>Im(T_{xy})</i>	<i>Re(T_{yx})</i>	<i>Im(T_{yx})</i>	<i>Re(T_{yy})</i>	<i>Im(T_{yy})</i>
	<i>SE of $Re(T_{xx})$</i>	<i>SE of $Im(T_{xx})$</i>	<i>SE of $Re(T_{xy})$</i>	<i>SE of $Im(T_{xy})$</i>				
			<i>SE of $Re(T_{yx})$</i>	<i>SE of $Im(T_{yx})$</i>	<i>SE of $Re(T_{yy})$</i>	<i>SE of $Im(T_{yy})$</i>		

If SE is negative, the corresponding component is not used in the inversion.

Phase tensor (1/2)

PT	Number of observation stations of phase tensor (N_{pt})				
Station ID	Station ID for horizontal magnetic field	Type of the owner element ^{*1)}	Type of the electric field ^{*2)}	X coordinate value of the station (km)	Y coordinate value of the station (km)
Number of frequencies (N_f)					
Observed data of the 1 st frequency ^{*3)}					
⋮					
Observed data of the N_f -th frequency ^{*3)}					
					Data for the 1 st station of phase tensor
⋮					
					Data for the N_{pt} -th station of phase tensor

*1) Type of the owner element (0: Downward element, 1: Upward element) is needed only if the option of the keyword 'OWNER_ELEMENT' in 'control.dat' is -1.

*2) Type of the electric field (0: Horizontal electric field, 1: Tangential electric field) is needed only if the option of the keyword 'ELEC_FIELD' in 'control.dat' is -1.

*3) The format of the observed data of each frequency is described in the next slide

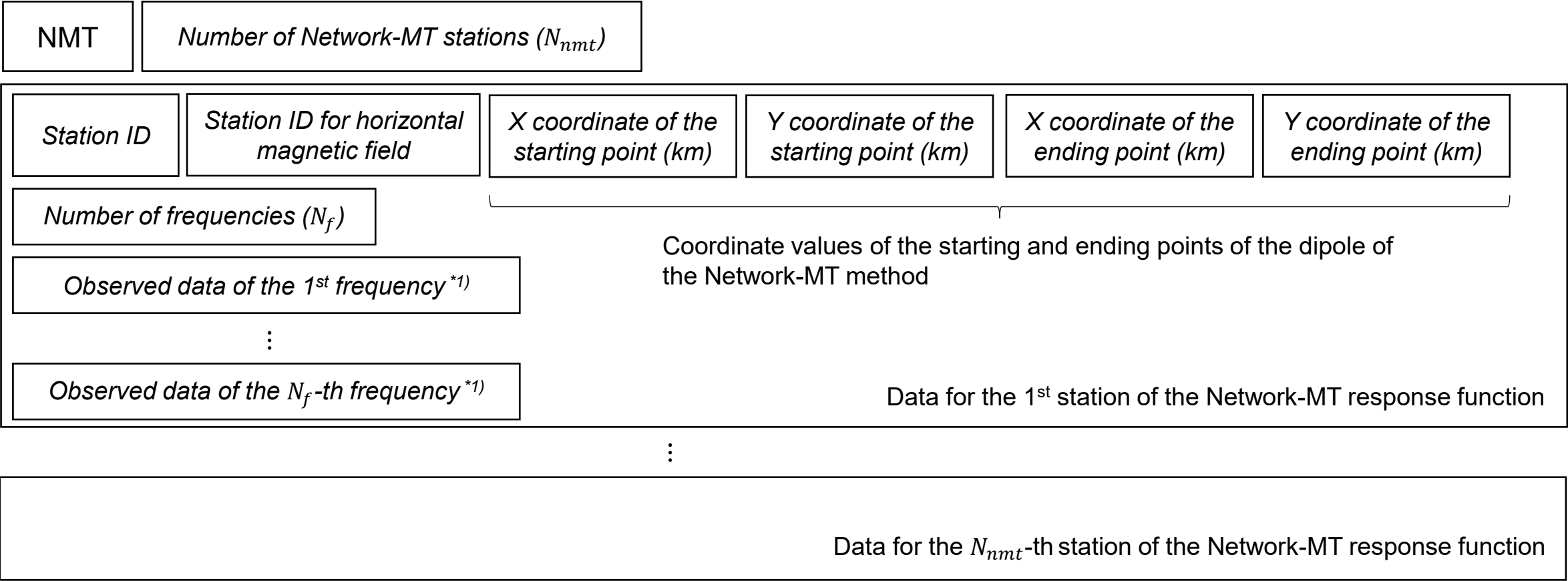
Phase tensor (2/2)

The format of the observed data of each frequency

<i>Frequency(Hz)</i>	Φ_{11}	Φ_{12}	Φ_{21}	Φ_{22}	<i>SE of Re(Φ_{11})</i>	<i>SE of Im(Φ_{12})</i>	<i>SE of Re(Φ_{21})</i>	<i>SE of Im(Φ_{22})</i>
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If SE is negative, the corresponding component is not used in the inversion.

Network-MT response function (1/2)



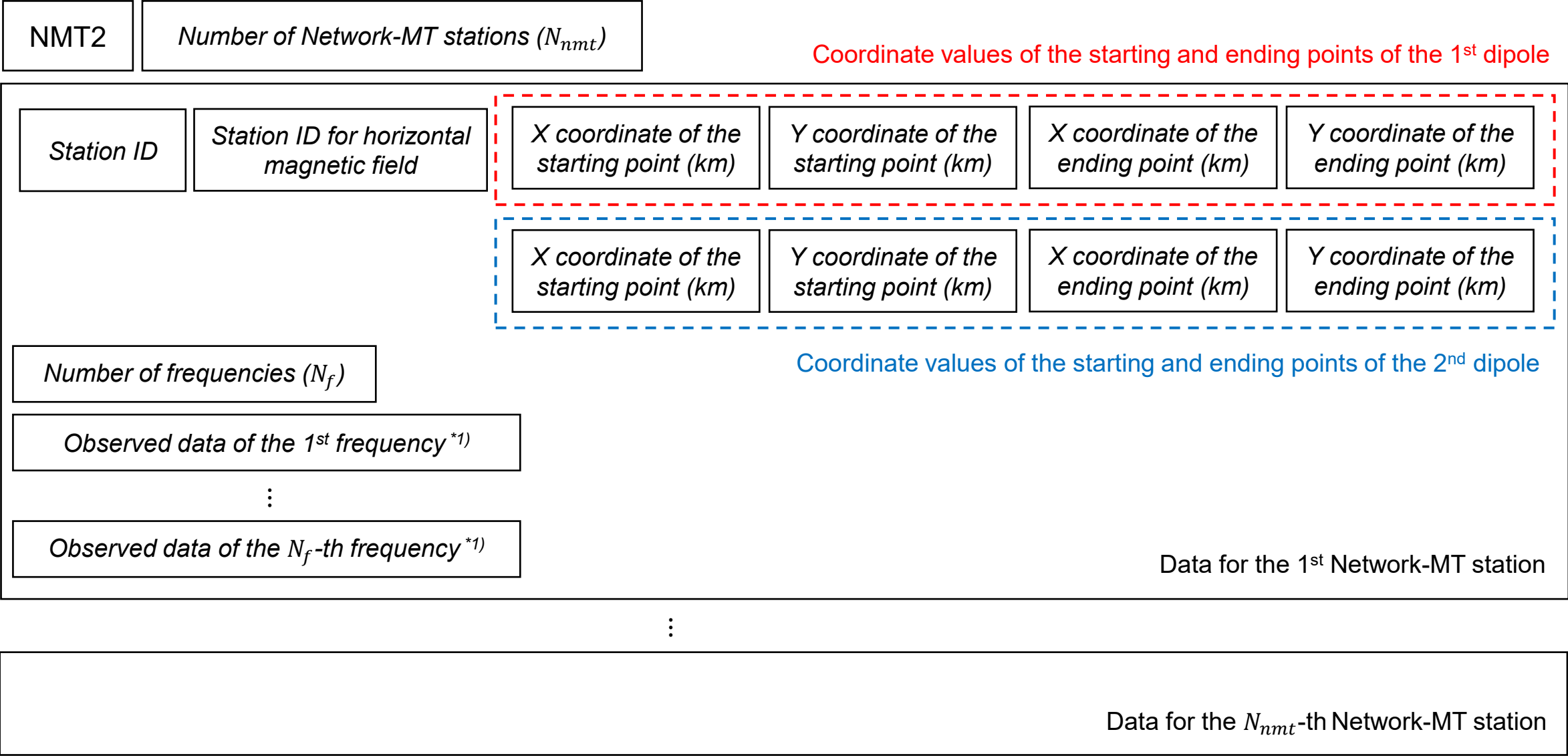
*1) The format of the observed data of each frequency is described in the next slide

Network-MT response function (2/2)

The format of the observed data of each frequency

<i>Frequency(Hz)</i>	<i>Re(V_x) (Vm/A)</i>	<i>Im(V_x) (Vm/A)</i>	<i>Re(V_y) (Vm/A)</i>	<i>Im(V_y) (Vm/A)</i>	<i>SE of Re(V_x)</i>	<i>SE of Im(V_x)</i>	<i>SE of Re(V_y)</i>	<i>SE of Im(V_y)</i>
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Impedance tensor obtained from the NMT response functions of two dipoles (1/2)



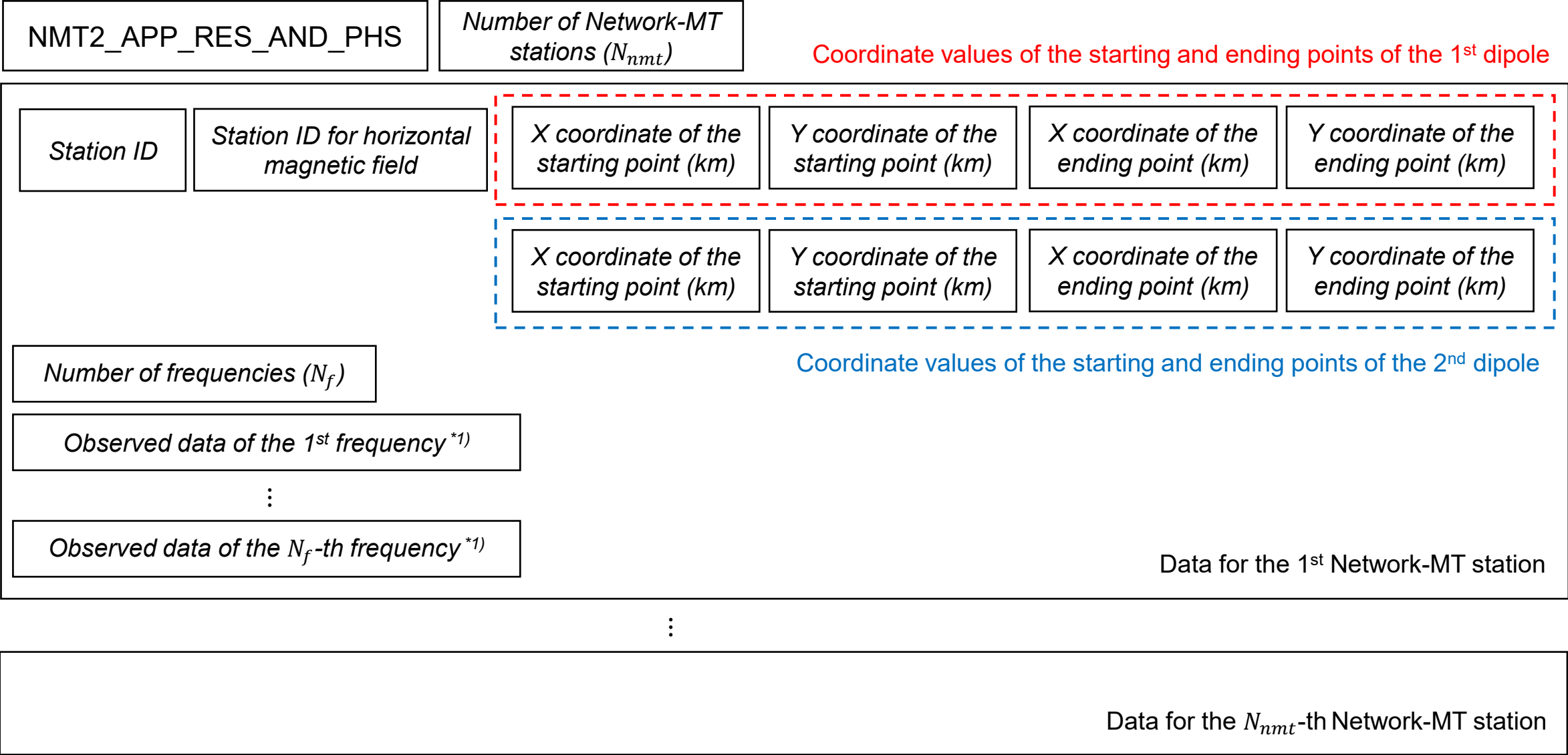
*1) The format of the observed data of each frequency is described in the next slide

Impedance tensor obtained from the NMT response functions of two dipoles (1/2)

The format of the observed data of each frequency

<i>Frequency(Hz)</i>	<i>Re(Z_{xx})</i> (V/A)	<i>Im(Z_{xx})</i> (V/A)	<i>Re(Z_{xy})</i> (V/A)	<i>Im(Z_{xy})</i> (V/A)	<i>Re(Z_{yx})</i> (V/A)	<i>Im(Z_{yx})</i> (V/A)	<i>Re(Z_{yy})</i> (V/A)	<i>Im(Z_{yy})</i> (V/A)
	<i>SE of Re(Z_{xx})</i>	<i>SE of Im(Z_{xx})</i>	<i>SE of Re(Z_{xy})</i>	<i>SE of Im(Z_{xy})</i>				
			<i>SE of Re(Z_{yx})</i>	<i>SE of Im(Z_{yx})</i>	<i>SE of Re(Z_{yy})</i>	<i>SE of Im(Z_{yy})</i>		

Apparent resistivity and phase obtained from the NMT response functions of two dipoles (1/2)



*1) The format of the observed data of each frequency is described in the next slide

Apparent resistivity and phase obtained from the NMT response functions of two dipoles (1/2)

The format of the observed data of each frequency

<i>Frequency(Hz)</i>	<i>$\rho_{a,xx}$ (Ωm)</i>	<i>θ_{xx} (<i>deg.</i>)</i>	<i>$\rho_{a,xy}$ (Ωm)</i>	<i>θ_{xy} (<i>deg.</i>)</i>	<i>$\rho_{a,yx}$ (Ωm)</i>	<i>θ_{yx} (<i>deg.</i>)</i>	<i>$\rho_{a,yy}$ (Ωm)</i>	<i>θ_{yy} (<i>deg.</i>)</i>
	<i>SE of $\rho_{a,xx}$</i>	<i>SE of θ_{xx}</i>	<i>SE of $\rho_{a,xy}$</i>	<i>SE of θ_{xy}</i>				
		<i>SE of $\rho_{a,yx}$</i>	<i>SE of θ_{yx}</i>	<i>SE of $\rho_{a,yy}$</i>	<i>SE of θ_{yy}</i>			

If SE is negative, the corresponding component is not used in the inversion.

Indication of the end of observation data

Impedance tensor

Apparent resistivity & phase

Vertical magnetic transfer function

Horizontal magnetic transfer function

Phase tensor

Network-MT response function

Impedance tensor obtained from the NMT response functions of two dipoles

Apparent resistivity and phase obtained from the NMT response functions of two dipoles

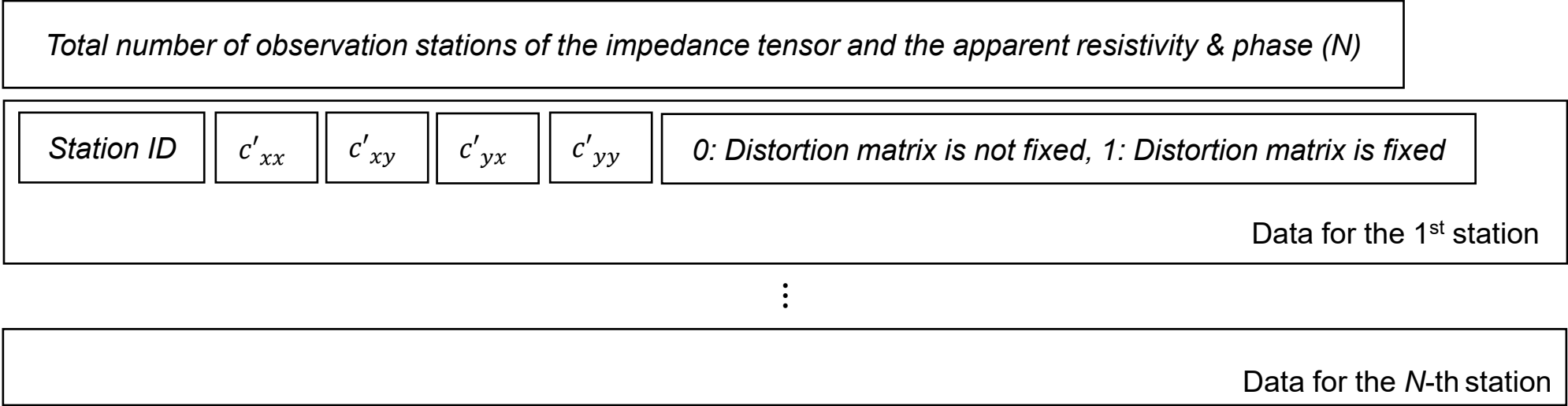
- You can write those response functions in no particular order.
- It is not necessary to write the information of all types of response functions.
- If you do not use some response functions, what you need to do is just not writing the keywords of the corresponding response functions (e.g., MT, VTF, and HTF).

END

Keyword 'END' must be written at the end of observation data

File format of distortion_iter[Iter#].dat (1/3)

When keyword 'DISTORTION' is 1 (differences of distortion matrix and unit matrix are estimated),



Distortion matrix: $\mathbf{Z}_{observed} = \begin{pmatrix} c_{xx} & c_{xy} \\ c_{yx} & c_{yy} \end{pmatrix} \mathbf{Z}_{regional}$

$\begin{pmatrix} c'_{xx} & c'_{xy} \\ c'_{yx} & c'_{yy} \end{pmatrix} = \begin{pmatrix} c_{xx} & c_{xy} \\ c_{yx} & c_{yy} \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in \mathbb{R}^{2 \times 2}$

Model parameters

File format of distortion_iter[Iter#].dat (2/3)

When keyword 'DISTORTION' is 2 (both gains and rotations are estimated),

Total number of observation stations of the impedance tensor and the apparent resistivity & phase (N)					
Station ID	G _x	G _y	β _x (deg.)	β _y (deg.)	0: Distortion matrix is not fixed, 1: Distortion matrix is fixed
					Data for the 1 st station
⋮					
					Data for the N-th station

Distortion matrix is decomposed as Smith (1995)

Distortion matrix:
$$\mathbf{Z}_{observed} = \begin{pmatrix} 10^{G_x} \begin{pmatrix} \cos\beta_x \\ \sin\beta_x \end{pmatrix} & 10^{G_y} \begin{pmatrix} -\sin\beta_y \\ \cos\beta_y \end{pmatrix} \end{pmatrix} \mathbf{Z}_{regional}$$

Smith, J. T. (1995). Understanding telluric distortion matrices. *Geophysical Journal International*, 122(1), 219–226. <https://doi.org/10.1111/j.1365-246X.1995.tb03549.x>

File format of distortion_iter[Iter#].dat (3/3)

When keyword 'DISTORTION' is 3 (only gains are only estimated),

Total number of observation stations of the impedance tensor and the apparent resistivity & phase (N)			
Station ID	G_x	G_y	0: Distortion matrix is not fixed, 1: Distortion matrix is fixed
			Data for the 1 st station
⋮			
			Data for the N-th station

Distortion matrix: $\mathbf{Z}_{observed} = \begin{pmatrix} 10^{G_x} & 0 \\ 0 & 10^{G_y} \end{pmatrix} \mathbf{Z}_{regional}$

Output files

File name	Content
femtic_[PE#].log	Log information about calculation
femtic.cnv	Convergence of Gauss-Newton iteration
result_[PE#]_iter[Iter#].csv	Calculated response functions and observed response functions
resistivity_block_iter[Iter#].dat	Resistivity values obtained by the inversion
distortion_iter[Iter#].dat	Parameters of galvanic distortion obtained by the inversion

In the file names, [PE#] and [Iter#] indicate process number and iteration number, respectively.

Output files for the visualization with ParaView (1/4)

These output files can be read and visualized by ParaView.

File name	Content
obs_loc.vtk	Locations of the observation stations used in the inversion
induction_arrow_[PE#]_iter[Iter#].vtk	Induction vector (Parkinson convention) at each station of the vertical magnetic transfer function
result.case	EnSight Gold case file
Mesh.geo	EnSight Gold geometry file
BlockIDs	ID of parameter cells (EnSight Gold variable file format)
AnisotropyTypes	Type of anisotropy (0: isotropy, 1: transverse isotropy, 2: general anisotropy)
RhoXX.iter[Iter#]	Value of ρ_{xx} (Ωm) (EnSight Gold variable file format)
RhoYY.iter[Iter#]	Value of ρ_{yy} (Ωm) (EnSight Gold variable file format)
RhoZZ.iter[Iter#]	Value of ρ_{zz} (Ωm) (EnSight Gold variable file format)
Strike.iter[Iter#]	Value of θ_s (deg.) (EnSight Gold variable file format)
Dip.iter[Iter#]	Value of θ_D (deg.) (EnSight Gold variable file format)
Slant.iter[Iter#]	Value of θ_L (deg.) (EnSight Gold variable file format)

*1) [PE#] and [Iter#] indicate process number and iteration number, respectively.

Output files for the visualization with ParaView (2/4)

File name	Content
Anisotropy.iter[Iter#]	$ \log(\rho_{xx}) - \log(\rho_{yy}) $ (EnSight Gold variable file format)
ReE_Freq[Freq]_ExPol.iter[Iter#]	Real part of the electric field for the Ex-polarization (EnSight Gold variable file format)
ImE_Freq[Freq]_ExPol.iter[Iter#]	Imaginary part of the electric field for the Ex-polarization (EnSight Gold variable file format)
ReE_Freq[Freq]_EyPol.iter[Iter#]	Real part of the electric field for the Ey-polarization (EnSight Gold variable file format)
ImE_Freq[Freq]_EyPol.iter[Iter#]	Imaginary part of the electric field for the Ey-polarization (EnSight Gold variable file format)
ReH_Freq[Freq]_ExPol.iter[Iter#]	Real part of the magnetic field for the Ex-polarization (EnSight Gold variable file format)
ImH_Freq[Freq]_ExPol.iter[Iter#]	Imaginary part of the magnetic field for the Ex-polarization (EnSight Gold variable file format)
ReH_Freq[Freq]_EyPol.iter[Iter#]	Real part of the magnetic field for the Ey-polarization (EnSight Gold variable file format)
ImH_Freq[Freq]_EyPol.iter[Iter#]	Imaginary part of the magnetic field for the Ey-polarization (EnSight Gold variable file format)

*1) [Freq] and [Iter#] indicate frequency and iteration number, respectively.

Output files for the visualization with ParaView (3/4)

File name	Content
Rej_Freq[Freq]_ExPol.iter[Iter#]	Real part of the electric current density for the Ex-polarization (EnSight Gold variable file format)
Imj_Freq[Freq]_ExPol.iter[Iter#]	Imaginary part of the electric current density for the Ex-polarization (EnSight Gold variable file format)
Rej_Freq[Freq]_EyPol.iter[Iter#]	Real part of the electric current density for the Ey-polarization (EnSight Gold variable file format)
Imj_Freq[Freq]_EyPol.iter[Iter#]	Imaginary part of the electric current density for the Ey-polarization (EnSight Gold variable file format)
Sensitivity.iter[Iter#]	Sensitivity (EnSight Gold variable file format) calculated as $s_i = \sum_j^{N_d} \left \frac{1}{\sigma_j} \frac{\partial d_j}{\partial \log(\rho_i)} \right $ N_d : Number of the observation data *2) d_j : The j-th data σ_j : Standard deviation of the j-th data ρ_i : Resistivity at the i-th parameter cell
NormalizedSensitivity.iter[Iter#]	Normalized sensitivity (EnSight Gold variable file format) calculated as $\bar{s}_i = s_i / \max_i s_i$

*1) [Freq] and [Iter#] indicate frequency and iteration number, respectively.

*2) Real part and imaginary part of each response function are treated as different data.

Output files for the visualization with ParaView (4/4)

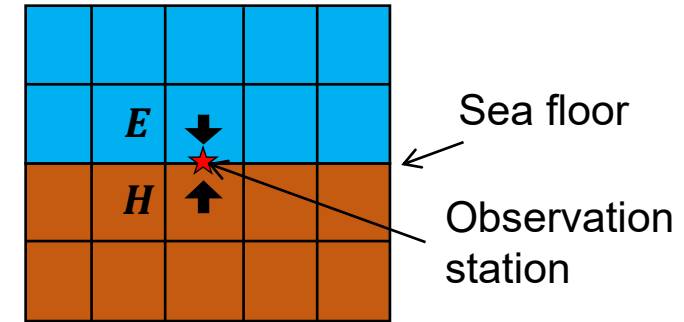
File name	Content
SensitivityDensity.iter[Iter#]	Sensitivity density (EnSight Gold variable file format) calculated as $s'_i = \frac{1}{V_i} \sum_j^{N_d} \left \frac{1}{\sigma_j} \frac{\partial d_j}{\partial \log(\rho_i)} \right $ N_d : Number of the observation data *2) d_j : The j-th data σ_j : Standard deviation of the j-th data ρ_i : Resistivity at the i-th parameter cell V_i : Volume of the i-th parameter cell
NormalizedSensitivityDensity.iter[Iter#]	Normalized sensitivity (EnSight Gold variable file format) calculated as $\overline{s'_i} = s'_i / \max_i s'_i$
result_[PE#]_iter[Iter#].vtk	Output variables specified in 'OUTPUT_PARAM' (VTK file format (ASCII))

*1) [PE#] and [Iter#] indicate process number and iteration number, respectively.

*2) Real part and imaginary part of each response function are treated as different data.

Things to be careful when seafloor MT data is used

- For seafloor MT stations, it is better to interpolate the electric field and the magnetic field from the upper-side and lower-side elements, respectively (Usui et al., 2018).
- However, the use of different elements for interpolating the electric field and the magnetic field needs a bit tricky setting.
- The required setting is as follows.



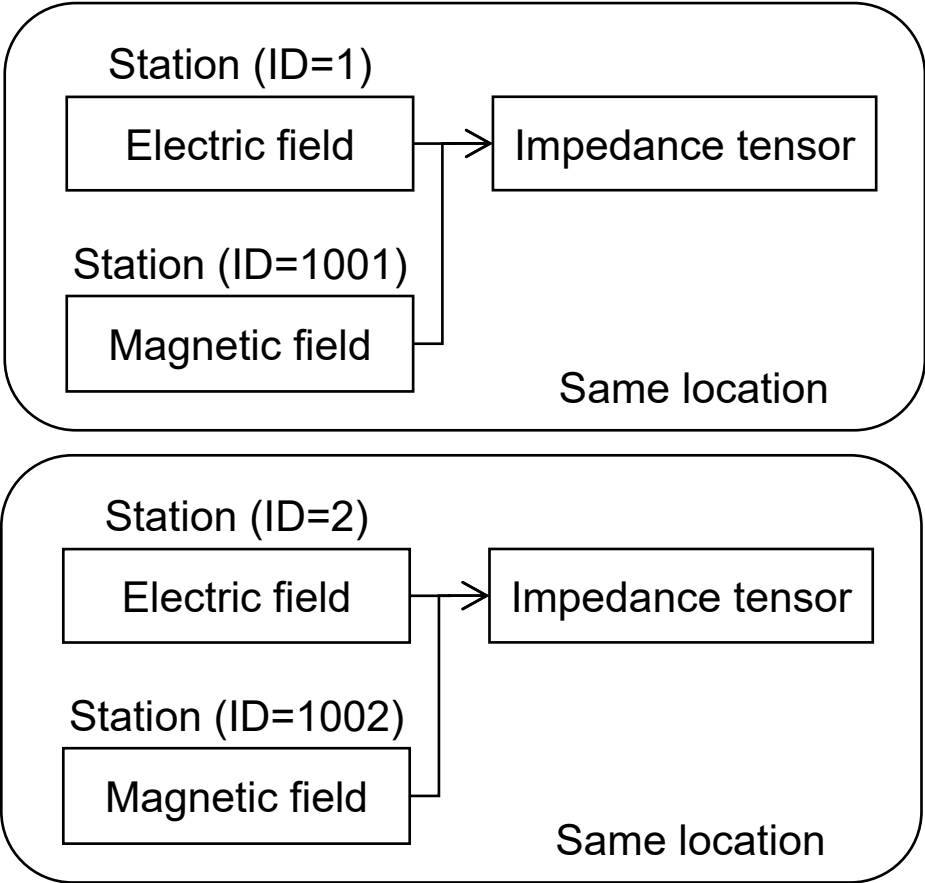
1. Set the option of the keyword 'OWNER_ELEMENT' of 'control.dat' to -1. By this setting, type of owner element can be individually selected for each element.
2. In 'observe.dat', add the dummy stations of the vertical magnetic transfer function or the horizontal magnetic transfer function at the same locations as seafloor MT stations. (If actually there are stations of the vertical magnetic transfer function or the horizontal magnetic transfer function on the same locations as the MT stations, you need not to add new dummy stations.)
3. For the dummy stations of magnetic transfer functions in 'observe.dat', set 'Type of the owner element' to 0 (element just below the sea floor is used for calculating the magnetic field)
4. For the seafloor MT stations in 'observe.dat', set 'Type of the owner element' to 1 (element just above the sea floor is used for calculating the electric field) and set 'Station ID for horizontal magnetic field' to the ID of the dummy station at the same location.

Things to be careful when seafloor MT data is used: Example

MT 2
1 1001 1 1.020000E+00 2.005000E+00
12
[Observed data of the 1st frequency]
...
[Observed data of the 12-th frequency]
2 1002 1 1.003330E+01 -9.004900E+01
12
[Observed data of the 1st frequency]
...
[Observed data of the 12-th frequency]

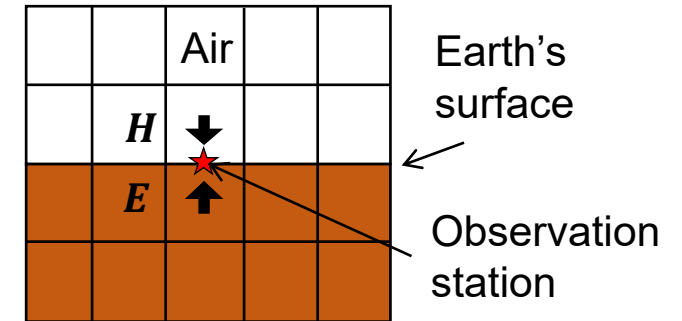
VTF 2
1001 1001 0 1.020000E+00 2.005000E+00
0
1002 1002 0 1.003330E+01 -9.004900E+01
0

} Dummy stations



How to improve response functions at high frequencies

- For land MT stations, it is better to interpolate the electric field and the magnetic field from the lower-side and upper-side elements, respectively.
- This is because the change of the magnetic field is smaller in the upper-side (in the air).
- Otherwise, it is possible that the amplitude of the impedance tensor is overestimated unless the size of the element just below MT station is not sufficiently small.
- The recommended setting is as follows.



1. Set the option of the keyword 'OWNER_ELEMENT' of 'control.dat' to -1. By this setting, type of owner element can be individually selected for each element.
2. In 'observe.dat', add the dummy stations of the vertical magnetic transfer function or the horizontal magnetic transfer function at the same locations as land MT stations. (If actually there are stations of the vertical magnetic transfer function or the horizontal magnetic transfer function on the same locations as the MT stations, you need not to add new dummy stations.)
3. For the dummy stations of magnetic transfer functions in 'observe.dat', set 'Type of the owner element' to 1 (element just above the Earth's surface is used for calculating the magnetic field)
4. For the land MT stations in 'observe.dat', set 'Type of the owner element' to 0 (element just below the Earth's surface is used for calculating the electric field) and set 'Station ID for horizontal magnetic field' to the ID of the dummy station at the same location.

How to improve response functions at high frequencies: Example

```
MT 2
1 1001 0 1.020000E+00 2.005000E+00
12
[Observed data of the 1st frequency]
...
[Observed data of the 12-th frequency]
2 1002 0 1.003330E+01 -9.004900E+01
12
[Observed data of the 1st frequency]
...
[Observed data of the 12-th frequency]
```

```
VTF 2
1001 1001 1 1.020000E+00 2.005000E+00
0
1002 1002 1 1.003330E+01 -9.004900E+01
0
```

} Dummy stations

